

ASSIGNMENT NO. 8

PROBLEM STATEMENT:

Write a program in C to accept marks of five courses of a student and compute the result. A student is considered PASS if he/she scores 40 marks or more in each course. If the student passes, calculate the aggregate percentage and assign the grade as follows:

- Aggregate $\geq 75\%$: Distinction
- Aggregate $\geq 60\%$ and $< 75\%$: First Division
- Aggregate $\geq 50\%$ and $< 60\%$: Second Division
- Aggregate $\geq 40\%$ and $< 50\%$: Third Division

OBJECTIVE:

- To understand the use of conditional statements in C
- To apply logical operators for decision making
- To calculate total, percentage, and grade of a student
- To enhance problem-solving skills using real-life scenarios

THEORY:

In academic evaluation systems, student performance is assessed based on individual subject marks as well as overall aggregate performance. In C programming, such evaluation systems can be implemented using conditional statements like if, else if, and else. Logical operators help in checking multiple conditions simultaneously. Percentage calculation involves arithmetic operations, and grading logic is implemented based on predefined ranges. This program demonstrates structured programming and decision-making in C.

CODE:

```
#include <stdio.h>

int main() {
    int marks[5], i, total = 0;
    int pass = 1; // Assume pass initially
    float percentage;

    // Input marks for 5 courses
    printf("Enter marks for 5 courses (out of 100 each):\n");
    for(i = 0; i < 5; i++) {
        printf("Course %d: ", i+1);
        scanf("%d", &marks[i]);

        // Check if student failed in any course
        if(marks[i] < 40) {
            pass = 0;
        }
        total += marks[i];
    }

    // Display marks entered
    printf("\nMarks entered:\n");
    for(i = 0; i < 5; i++) {
        printf("Course %d: %d\n", i+1, marks[i]);
    }

    // Check result and compute grade
    printf("\n--- RESULT ---\n");
    if(pass == 0) {
        printf("RESULT: FAIL\n");
        printf("Student failed in at least one course.\n");
    } else {
        percentage = (float)total / 5;
        printf("RESULT: PASS\n");
        printf("Aggregate Percentage: %.2f%%\n", percentage);

        // Assign grade based on percentage
        if(percentage >= 75) {
            printf("GRADE: Distinction\n");
        } else if(percentage >= 60) {
            printf("GRADE: First Division\n");
        } else if(percentage >= 50) {
            printf("GRADE: Second Division\n");
        } else if(percentage >= 40) {
            printf("GRADE: Third Division\n");
        }
    }

    return 0;
}
```

OUTPUT:

```
Enter marks for 5 courses (out of 100 each):  
Course 1: 99  
Course 2: 80  
Course 3: 98  
Course 4: 88  
Course 5: 100  
  
Marks entered:  
Course 1: 99  
Course 2: 80  
Course 3: 98  
Course 4: 88  
Course 5: 100  
  
--- RESULT ---  
RESULT: PASS  
Aggregate Percentage: 93.00%  
GRADE: Distinction
```

CONCLUSION:

Thus, the program to calculate student result and grade based on marks was successfully implemented using C programming. This program demonstrates the effective use of conditional statements, logical operators, and arithmetic operations.